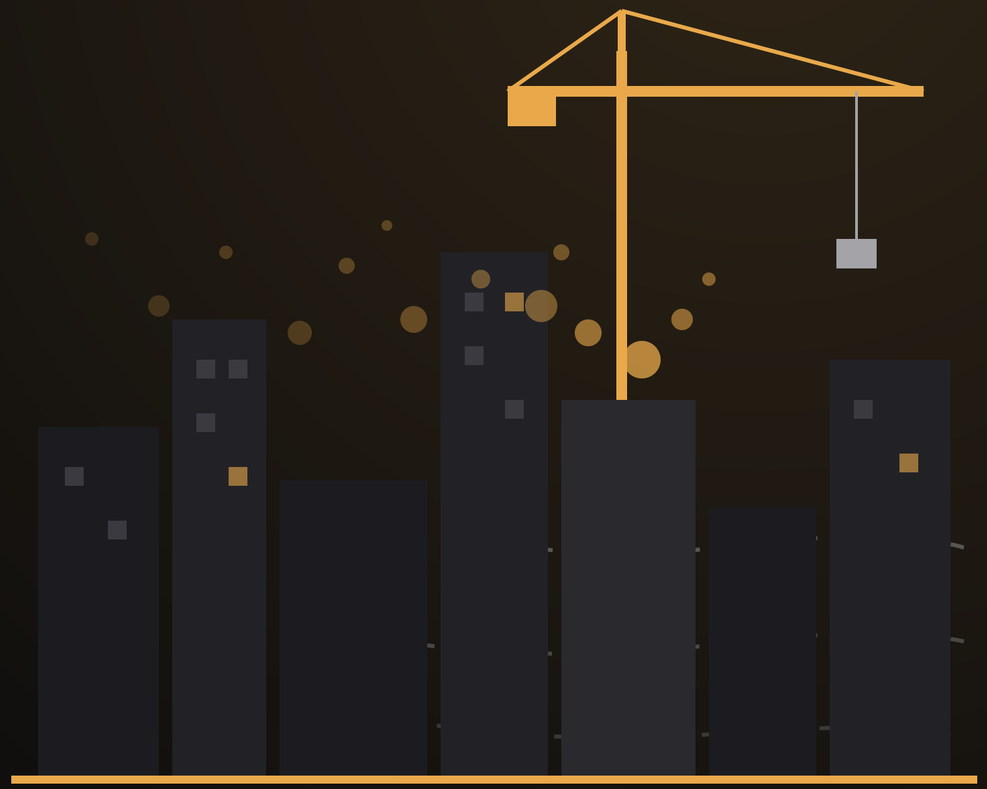
Techfest, IIT Bombay · InnovateX 2026–27 · Theme 2

DustTrace

Stop construction dust before it reaches people.

We forecast Mumbai's dust spikes 48 hours ahead, fix the cause at each site, capture the dust, and point BMC at the sites to inspect.



Team ARA IGNITE · Ansh, Dhruv, Anuj, Sarang

The problem statement · Theme 2

Rising AQI in modern urban cities

The challenge, as set by Techfest

"How can technologies be leveraged to monitor, predict, and reduce rising AQI levels in modern urban environments? Propose an innovative solution using IoT sensors, AI-driven air quality forecasting, adaptive traffic control, drone-based monitoring, or urban planning strategies. Your solution should address a specific pollution source and outline its benefits, feasibility, and implementation strategy."

Monitor
Existing sensors + our own nodes

Predict
Spike forecast, 24–72 h ahead

Reduce
Reschedule, capture, clean, enforce

Our specific source

Construction dust

BMC's top-ranked source in Mumbai

Why we chose construction dust

Construction dust is Mumbai’s leading air-pollution source

~1,200
active construction sites across the city
Oct 2025

45%
had installed the mandated air-quality sensor
Oct 2025 · 535 sites

117
sensors found inactive in a 95-ward inspection
Nov 2025

62
stop-work notices in a single enforcement drive
Nov 2025

Since June 2025 every site must stream dust readings to BMC every 30 s. The city has the sensors, but nothing uses them to prevent a spike.

Sources: BMC Mumbai Air Pollution Mitigation Plan, via Karmactive (Oct 2025); Free Press Journal (27 Nov 2025)

Where existing solutions fall short

Four gaps between the sensors and enforcement

Untrusted data

Each sensor is owned, placed and maintained by the developer it monitors, and nobody checks it.

Blind spots

Homes, schools and many sites have no sensor at all; unregistered demolition never will.

Nothing prevents

Sprinklers and inspections start after AQI turns bad. No one tells a site when not to demolish.

No attribution

A CAAQMS station reads a whole neighbourhood. It cannot say which of dozens of upwind sites caused a spike.

Sources: Karmactive (Oct 2025); Free Press Journal (Nov 2025)

Our idea

Stop the dust before it reaches people.

Every tool in Mumbai today asks who polluted, after the air has already turned bad. DustTrace forecasts when and where dust will hit homes, schools and hospitals, and changes what happens on site before it does.

01

Predict

When and where dust will spike, and who is downwind

02 · UNIQUE

Prevent

Fix the cause, move dusty work, capture and clean the dust

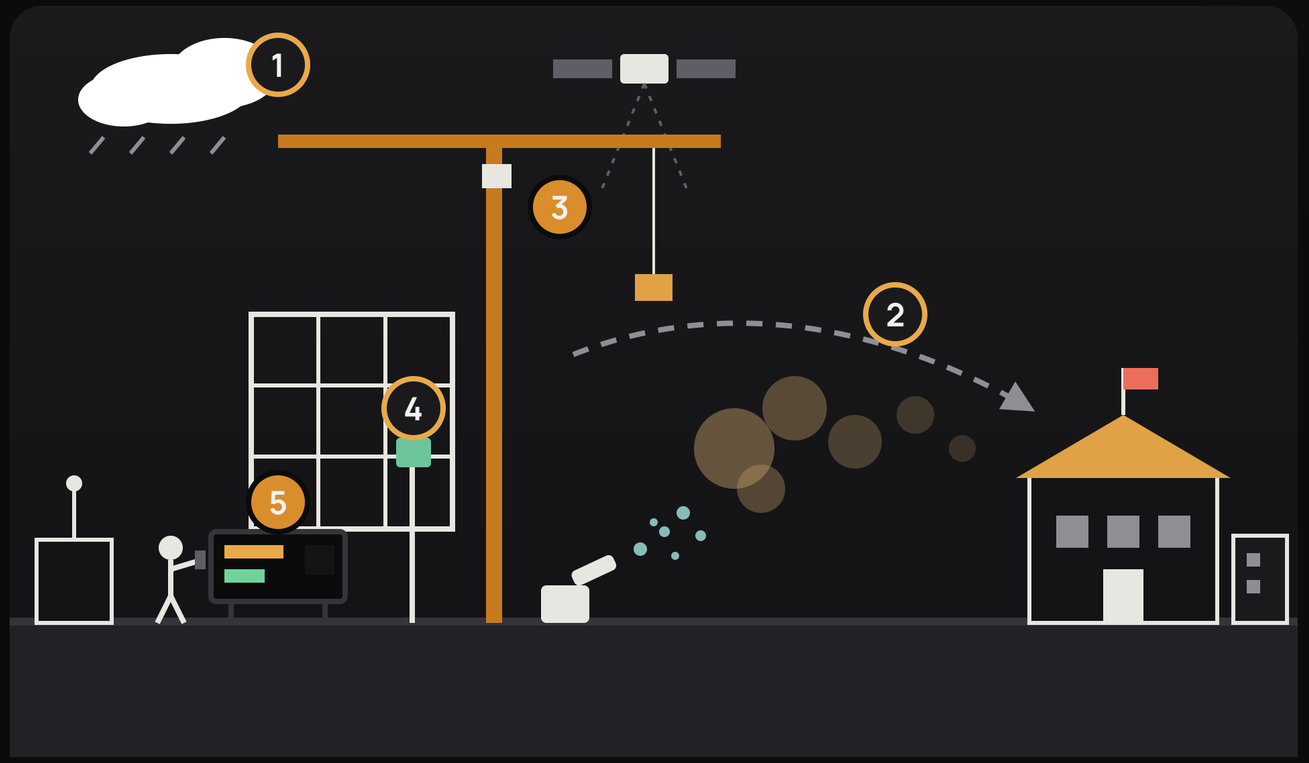
03

Prove

Audit sensors, find hidden sites, rank inspections, publish a score

Our solution

DustTrace on one Mumbai street



1 **Forecast · Predict**
Stations + weather forecast flag tomorrow's spike, ward by ward.

2 **Exposure map · Predict**
Plume model + our own node at the school show who breathes the dust.

3 **Safe-Work Window · Prevent · unique**
Dusty work moves to safe hours; a mist curtain captures dust first.

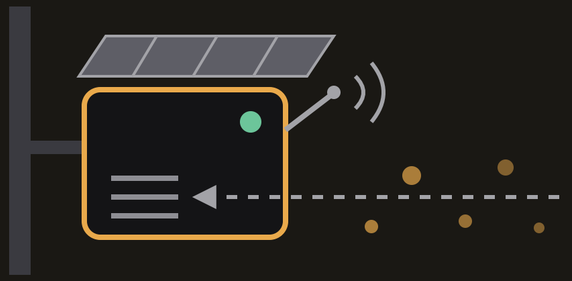
4 **Sensor Audit + Discover · Prove**
Site sensor checked against the station; satellite finds unsensored sites.

5 **Dust Score + QR · Prove · unique**
Score on the LED board; residents scan to report; officers get a ranked list.

Orange markers are our two unique elements. Runs on BMC site sensors and CPCB stations, plus low-cost DustTrace Nodes where no sensor exists.

Monitor · where no sensor exists

DustTrace Nodes: fixed and on BEST buses



ESP32 Wi-Fi board	₹450
PMS5003 laser dust sensor	₹1,400
Temp + humidity sensor	₹350
Enclosure, power, mount	₹500
Per node, vs ₹1–2 lakh site monitor	**≈ ₹2,700**

Where we install them

- **Fixed:** housing societies, schools and hospitals downwind
- **Fixed:** debris dumps and gaps between CPCB stations
- **Mobile:** on BEST buses with GPS, passing sites daily
- 2,792 BEST buses, 1,324 electric: power is easy

What they add

- What residents actually breathe
- Proof a Safe-Work Window worked, before vs after
- Bus readings at site gates check each site's own sensor
- Dust from sites with no sensor

Prices are team estimates. Nodes are calibrated against CPCB stations and used for screening; BMC site monitors stay the compliance record. BEST fleet: ThePrint, 2026.

Unique element 1 · Prevent

Safe-Work Windows: when dusty work is safe

Illustrative output · Site A, K-East · tomorrow, 07:00–19:00

07–09 · Avoid	09–11 · Avoid	11–13 · Caution	13–15 · Go	15–17 · Go	17–19 · Caution

07–11 Avoid: shallow morning mixing layer, light wind toward a school 400 m away

13–17 Go: sea breeze and a deeper mixing layer disperse dust; nothing sensitive downwind

How it's computed, per site, per hour

- Wind, mixing height and rain for the next 48 h
- Who is downwind: schools, hospitals, homes
- Tasks: demolition, excavation, cutting, loading

Why it works

Same work, better hour. No new equipment, no work lost, just less dust landing on people.

Measured in Round 2 as person-exposure avoided.

Prevent · beyond rescheduling

Fix the cause, then capture the dust

Cause	How DustTrace detects it	Fix sent to the site
Dry excavation, demolition	Work-hour spike of coarse dust	**Spray water; move to Safe-Work Window**
Truck track-out	Spikes at the gate at truck times; QR photos	**Wheel wash, covered loads, gate sweeping**
Uncovered debris	Satellite bare soil + photo classifier	**Cover with green net within 2 hours**
Dry, windy day ahead	Weather forecast: wind, no rain	**Pre-wet the site; pause dusty tasks**

Capture and clean, not just divert

Mist curtain

Fine mist along the site boundary, switched on by the forecast, catches dust before it leaves.

Green screens

Vegetation barriers on site hoardings trap coarse dust at the edge of the site.

Clean air inside

A school downwind in an Avoid window is told to close windows and run classroom purifiers.

Fixes follow BMC's 28-point dust guidelines, which already include water spraying, covered transport and green screens (MCGM 2023; Hindustan Times, 29 Mar 2023).

How actions are decided · Dust Response Agent

Every trigger becomes a verified fix

1 · Trigger
Forecast, sensor, node, QR, satellite

2 · Investigate
Sensors, wind, imagery, past cases

3 · Attribute
Competing causes, with odds

4 · Decide
Dust rules D1–D10

5 · Approve
Auto, site, L1, L2

6 · Act
Prevent, correct, enforce

7 · Verify
Did PM fall? If not, reopen

Rules that keep it fair

D1	No enforcement on a single signal; ask for a photo or spot check first
D7	No reply by the deadline: escalate. A reply is never assumed
D9	Stop-work is only ever recommended, with L2 human sign-off
D10	Not construction? Refer to MPCB; no action on the site

Who approves what

Auto	Safe-Work Windows, misting via relay, school alerts
Site	Wheel wash, covering, pausing dusty work
L1 officer	Inspection, notice, registering a hidden site
L2 AC	Stop-work recommendation only

Modelled on our fraud-investigation agent from TigerGraph Hacker House Goa. Runs on Postgres + PostGIS + pgvector. Rules D1–D10 are proposals to agree with BMC.

Unique element 2 · Prove

A Dust Score on every site’s LED board

SITE MH-KE-0231 · MOCK-UP

PM10 · µg/m³ **86**

PM2.5 · µg/m³ **41**

DUST SCORE

82

out of 100 · Good



Scan to report dust

What goes into the score

30%	Sensor trust, from the Audit
25%	Spikes traced back to the site
20%	Safe-Work Windows followed
15%	Verified citizen reports
10%	Inspection history

What it unlocks

80+ · fewer routine inspections

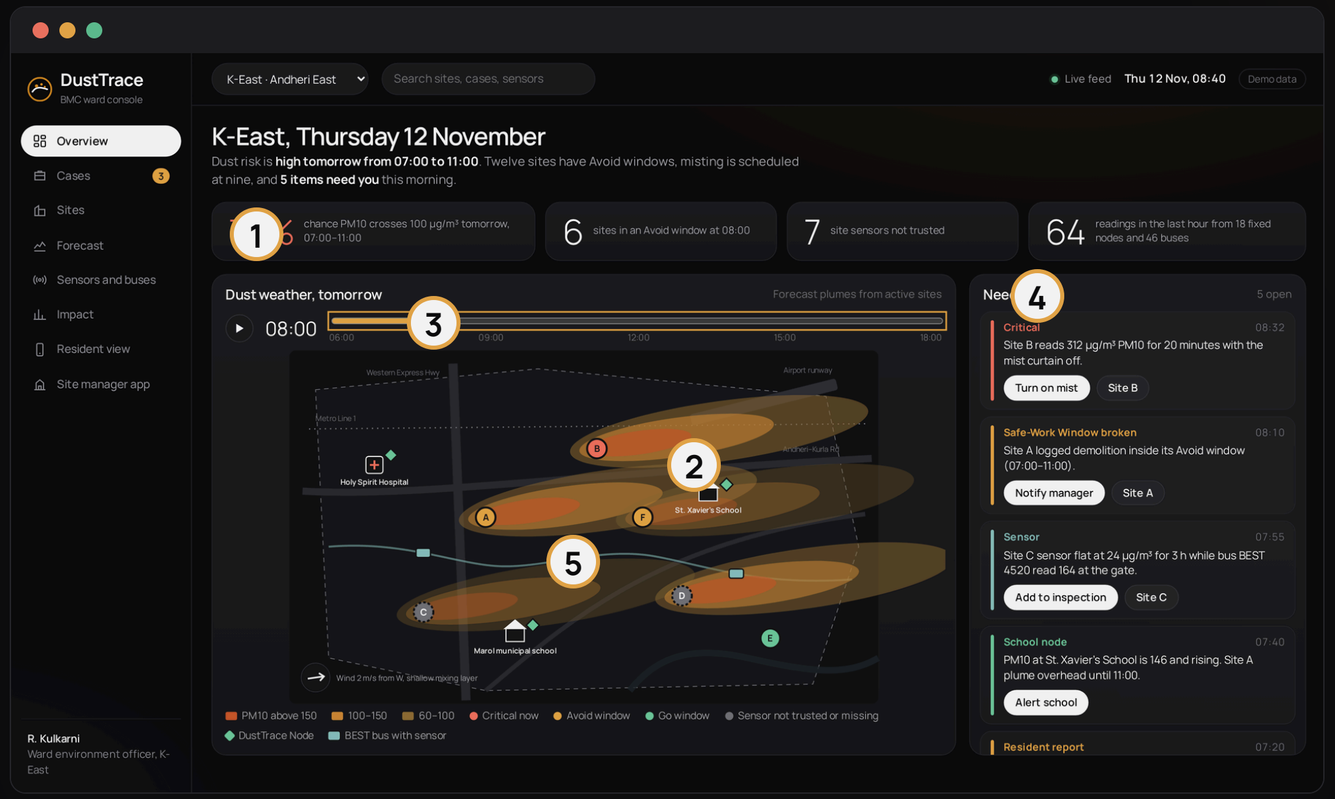
Below 40 · top of the inspection list

Public map · residents and homebuyers see it

The LED board is already mandatory at every site. Board layout is a mock-up; score weights and incentives are proposals to design with BMC.

Web prototype · ward control room

The ward officer’s morning, on one screen



1 **A morning briefing**
78% chance PM10 crosses 100 tomorrow, 07:00–11:00; 6 sites in Avoid windows; 7 sensors not trusted.

2 **Dust weather map**
Forecast plumes drift from each site with the wind, over schools and hospitals.

3 **Scrub through tomorrow**
Press play and watch plumes grow, shift and each site turn Go or Avoid.

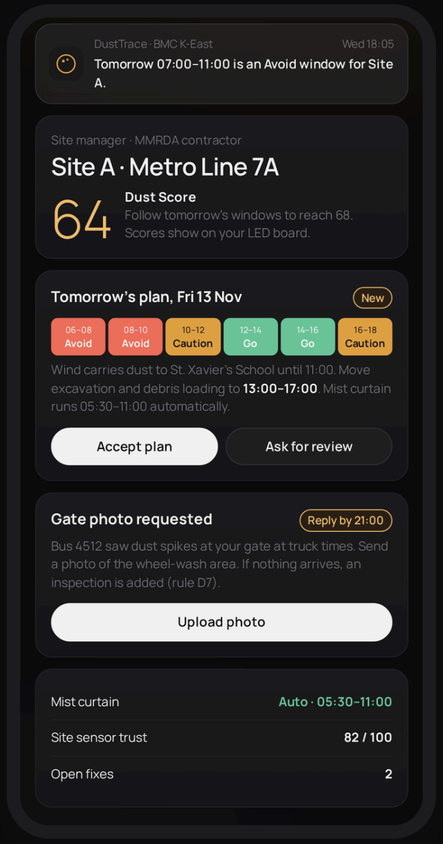
4 **Needs you**
Every alert carries its fix: turn on mist, notify the manager, add to inspection, alert the school.

5 **Our nodes and BEST buses**
Green: fixed DustTrace Nodes. Teal: buses carrying sensors past the sites.

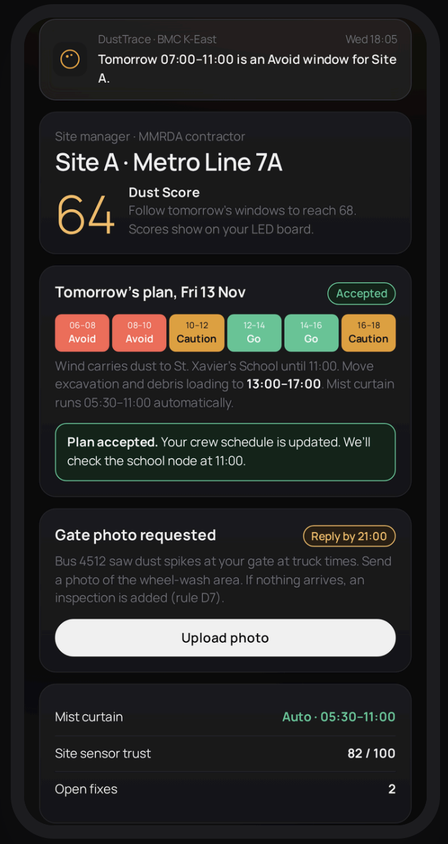
Screenshot of our clickable web prototype on demo data. Round 2 connects it to real station, weather and node data.

Web prototype · site manager app

The site gets a plan, not just a fine



18:05 · plan arrives



18:06 · one tap, accepted

The evening before, not the morning after

Tomorrow's Avoid hours and the better slot to move dusty work into, with the reason in one line.

One tap to accept

The crew schedule moves. The mist curtain is already set to run 05:30–11:00.

Evidence with a deadline

"Send a gate photo by 21:00." If nothing arrives, rule D7 adds an inspection; silence is never read as a yes.

A reason to comply

Following windows raises the Dust Score shown on the site's LED board, which residents can see.

Clickable prototype, demo data. A mobile web page: no app install for site staff.

Economic viability

Costs less than sensoring the blind spots

₹50–60 L

a year to run DustTrace city-wide: cloud, alerts and a 5-person team

Team estimate · + ~₹5 L for 200 nodes

₹5–10 Cr

to put sensors on ~500 unsensored sites at ₹1–2 lakh each, once

Plus 2-monthly calibration

₹938.59 Cr

air-quality grants released to Greater Mumbai, FY20–FY24

₹680.32 Cr utilised · our funding channel

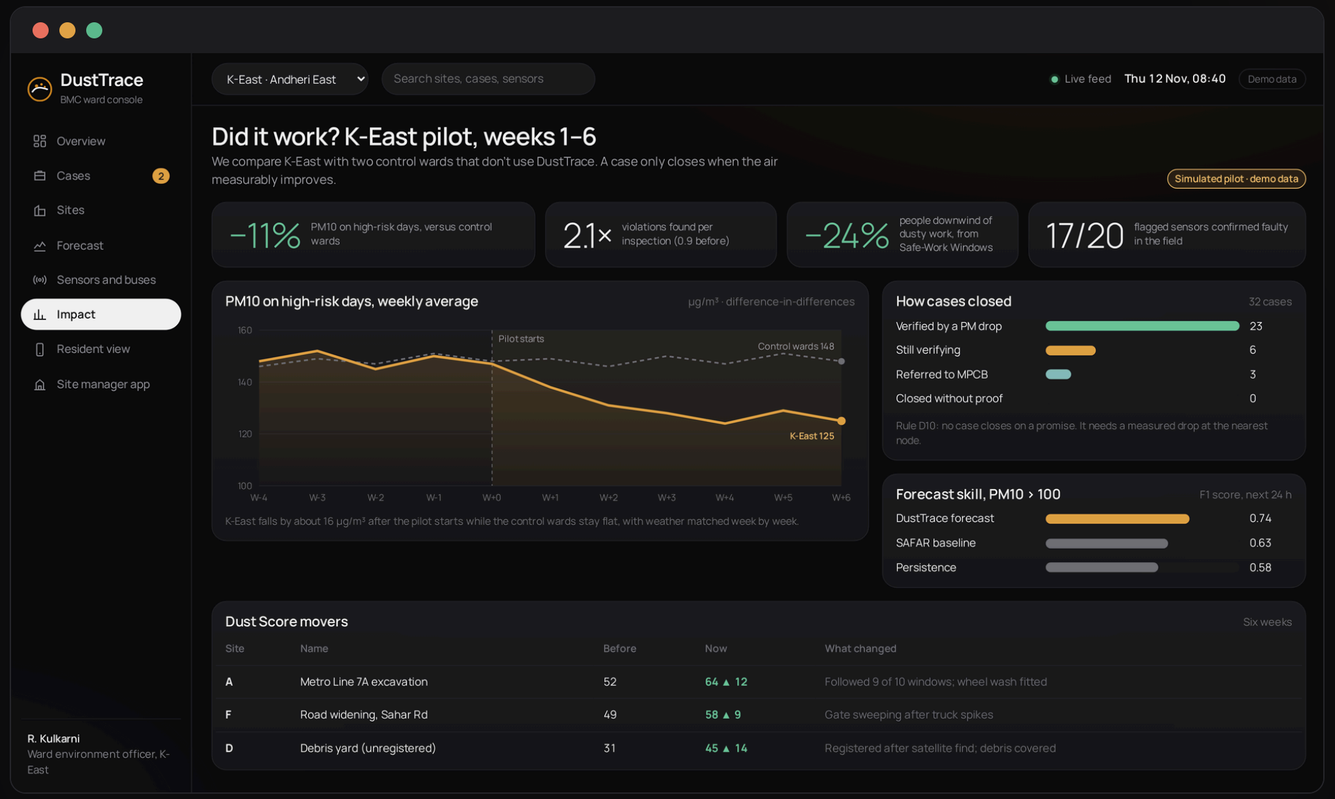
Revenue model · assumptions to validate

B2G licence to BMC at ₹2–3 lakh per ward a year · developer tier at ₹5,000 per site a year (Safe-Work Windows, misting schedule, Dust Score certificate) · data and API licences for MPCB and other cities

Sources: sensor cost, Karmactive (Oct 2025); NCAP / XV Finance Commission city releases, FY2019-20 to 2023-24 (MoEFCC data)

Impact

Proof, not promises



Pilot targets and how we measure them

≥ 2× violations per inspection
Inspection outcome logs

≥ 80% precision on faulty-sensor flags
Field checks of flags

≥ 70% recall on unsensored sites
Ward-office site survey

Forecast F1 ≥ 0.7, beats persistence
Back-test, 2022–26 data

−20% people downwind of dusty work
Plume model, real wind

−10% PM10 on high-risk days
Difference-in-differences vs control wards

Prototype screen on simulated demo data. Targets are our assumptions to validate, not results; health gains via WHO AirQ+.

Mumbai already paid for the sensors. DustTrace turns them into prevention.

In Round 2 we will show

A real past spike, and whether we forecast it 48 h ahead

Safe-Work Windows and the exposure they avoid, on real wind

Faulty sensors and unsensored sites, caught

Team ARA IGNITE · Ansh, Dhruv, Anuj, Sarang · Thank you

Appendix · The IoT already exists

Mumbai made site sensors compulsory in 2025

Oct 2023
BMC issues dust-mitigation guidelines for construction sites

Nov 2023
Bombay High Court takes suo motu cognizance (SMPIL 3/2023)

Jan 2025
Court orders site-level monitoring, replacing blanket ward bans

May 2025
BMC shortlists approved sensor models (circular MGC/F/6526)

Jun 2025
PM2.5/PM10 sensor and LED board mandatory at every active site

Nov 2025
44% developer compliance; 62 stop-work notices

662
site sensors installed (Nov 2025)

~400
actually linked to BMC's dashboard

30 s
refresh rate: PM2.5, PM10, temp, humidity

Sources: MCGM guidelines (25 Oct 2023); Karmactive (Oct 2025); Free Press Journal (Nov 2025); Tribune India

Appendix · Data sources

Built entirely on free, existing data

Data	Source	Access
PM10 / PM2.5 at 21+ Mumbai stations	CPCB CAAQMS via OpenAQ API	Free API key
Wind, humidity, rain, boundary layer	Open-Meteo	Free, no key
10 m satellite imagery, ~5-day revisit	Sentinel-2 via Earth Engine	Free, noncommercial
Registered construction projects	MahaRERA public registry	Public
Ward boundaries	DataMeet spatial data	Open licence
Site-sensor feed (30 s PM, humidity)	BMC Command & Control Centre	Simulated now; RTI + vendor data planned
Citizen photos	DustTrace app	Crowdsourced

₹0 in data costs. The one feed we lack is simulated from real station data.

Appendix · Architecture

From data to action, for three audiences

Data we reuse

- Site sensors (BMC)
- CPCB / SAFAR stations
- Wind + weather forecast
- Sentinel-2 imagery
- Citizen photos + OSM map

Engines

Forecast + exposure
Spikes 24–72 h; who is downwind

Sensor Audit
Trust score per sensor

Discover
Sites with no sensor

Outputs, by audience

Sites · Prevent
Safe-Work Windows + forecast-triggered misting

Officers · Prove
Ranked inspection list with reasons

Public · Prove
Dust Score on the LED board + QR reports

↻ Outcomes and citizen reports retrain every engine

Stack: Next.js · NestJS · Postgres + PostGIS + TimescaleDB + pgvector · Python (LightGBM, PyTorch) · Earth Engine

Appendix · Forecast and ranking models

Forecast the spike, then point at the sites

Will PM10 cross 100 µg/m³?

LightGBM per station on 2022–26 history, wind, humidity and upwind site density. Must beat persistence and SAFAR.

Priority score per site

upwind exposure + (1 − sensor trust) + citizen reports + no-sensor flag + past violations

Weights hand-tuned first, then learned from inspection outcomes.

Sample output · illustrative

K-East ward · inspect tomorrow

1 **Site A · Metro excavation**
Sensor flatlined 3 days · upwind of forecast spike

2 **Site B · demolition**
No sensor · 3 citizen photos this week

3 **Site C · redevelopment**
Reads cleaner than ambient while downwind

100 µg/m³ = India's 24-hour PM10 standard (NAAQS). Right panel is a mock-up of the officer view, not real data.

Appendix · What makes us different

What no existing tool does together

Capability	BMC dashboard	SAFAR	Vendor devices	DustTrace
Live PM per site	Yes	No	Yes	**Theirs + own nodes**
Dust spike forecast	No	City-level	No	**Ward-level, 72 h**
Checks sensor trust	No	No	No	**Yes, daily**
Finds unsensored sites	No	No	No	**Satellite + citizens**
When dusty work is safe	No	No	No	**Hourly, per site**
Mitigation before a spike	No	No	After threshold	**Forecast-triggered**
Ranked inspections	Threshold alerts	No	No	**Yes, with reasons**
Public site score	No	No	No	**Dust Score + QR**

Based on public descriptions of each tool as of Sep 2026. Vendor example: Oizom AQBot relay outputs trigger sprinklers when dust crosses a set limit.

Scalability

One source today, any fixed source next

Across cities

Tier 1 · Mumbai, Delhi-NCR
Site sensors mandated: the full product

Tier 2 · Pune, Bengaluru, Hyderabad, Chennai
Stations, no mandate: Predict, Prevent, Discover

Tier 3 · smaller NCAP cities
Few stations: satellite and citizen discovery

130
NCAP cities

₹16,424 Cr
grants released

Beyond dust, in Mumbai

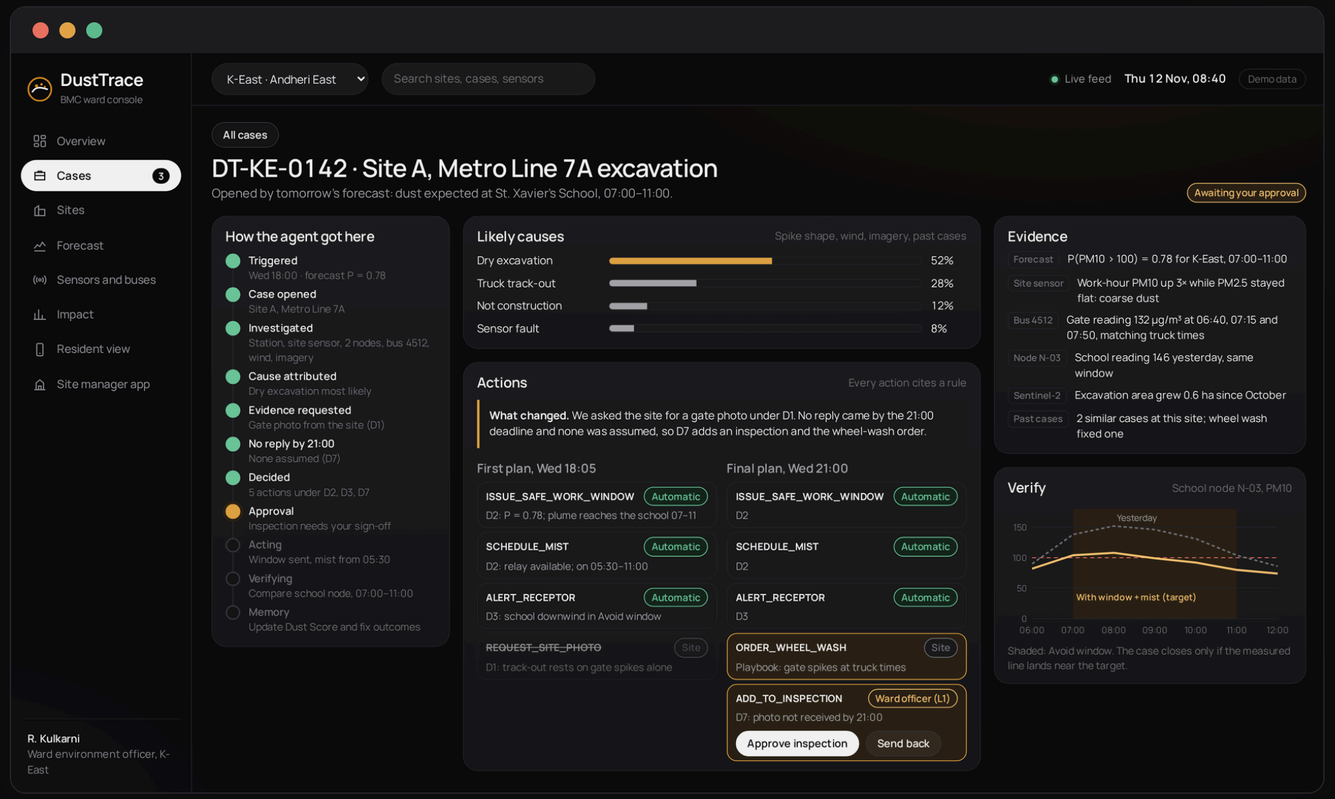
Next source	Why it fits	Mumbai signal
RMC plants	Fixed site, schedulable batching	Flagged to MPCB, Nov 2025
Wood-fired bakeries	Fixed chimney; Dust Score fits	Only 209 of 593 on clean fuel
Open waste burning	Citizen QR reports + hotspot map	₹5,000 fine planned by BMC
Road dust	Forecast drives sweeping and misting routes	730 roads in BMC's dust drive

Same engines: Forecast, Audit, Discover, Prioritise. Only the source profile changes.

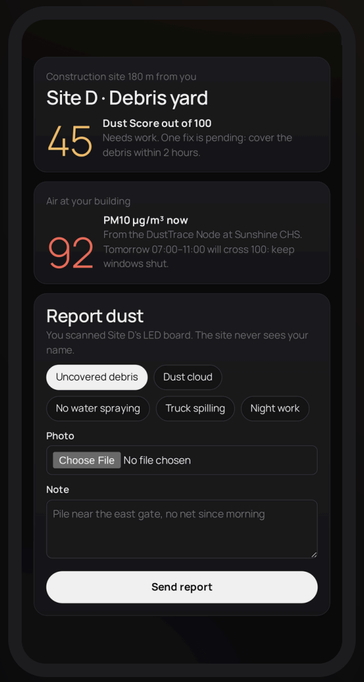
Sources: PIB, NCAP Parliament reply (₹16,423.56 Cr); Free Press Journal, 27 Nov 2025 (RMC plants, bakeries, roads, open-burning fine)

Web prototype · agent case and resident view

A case, decided and explained



Agent case: steps, causes, first vs final plan, evidence, then Verify.



Resident view, from the QR code

Try it live

- Approve the inspection
- Play tomorrow's dust map
- Turn on a site's mist
- Switch sites A–F
- File a QR report

Screenshots of our clickable web prototype, demo data. Every action names the rule that allowed it.